CSCI 218: Programming II (Spring 2024)

Class time: Tuesday/Thursday, 12:30 – 2:20pm (In-Person)

Location: Paine 212- Computer/Data Science classroom

Instructor	Babafemi Sorinolu ○ **Email:** babafemi.sorinolu@houghton.edu ○ **Office:** Paine 209B; 585.567.9259 ○ **Office hours:** Tuesday/Thursday, 2:30-4:30pm (Feel free to email me to schedule a virtual meeting.)
Moodle Page	https://moodle.houghton.edu/course/view.php?id=25626
Course Description	(4 credit hours) - Extends concepts learned in Programming I. Covers some advanced topics including object-oriented programming concepts using Java/Python, data structures, search algorithms, file handling, exceptions, threads, test-driven development, networking and sockets, and graphic user interface. **Prerequisite**: CSCI 211 or permission.
Required Textbooks	**Thinking in Java, 3rd Edition by Bruce Eckel**
Additional Textbook	**NA**
Required Software	Java, NetBeans or IntelliJ IDE, MySQL Workbench

Long-Range Goals:

1. The first of this course's main goals is to enhance your computer programming abilities and mastery of Object-Oriented Programming (OOP)

2. Develop a critical mindset for problem-solving through the application of algorithmic principles in programming to address and resolve challenges effectively.

3. Acquire fundamental Java skills for transitioning to advanced frameworks such as Spring, Java EE, Android development, and beyond.

Course Objectives: After successful completion of this course, students will be able to:

- Understand and apply advanced OOP concepts such as inheritance, polymorphism, encapsulation, and abstraction.
- Learn techniques for reading and writing files
- Implement multithreaded applications to achieve parallelism.
- Explore network programming concepts.
- Develop interactive applications with a graphical user interface.
- Foster collaboration in teams (pair programming)

Activities & Time Commitment: For a four-credit course, you should expect to commit at least 12 hours per week. Although not every week will be the same, these are typical weekly times:

- Class (4 hours): These would take the form of presentation-based instruction and discussions.
- Technical reading, study, & review (3 hours): There will be readings assigned each week. The student is expected to both read the text and take notes. This activity (along with the class presentation) will enhance the understanding and application of the concepts discussed during the course lectures.
- Homework (2 hours): Along with the readings, there will be assignments based on the assigned readings and class presentations. These assignments are designed to reinforce the understanding of key programming concepts through various exercises. **Work is due at the time it is assigned**, whether you are in class or not.
- Hands-on Activities and Quizzes (3 hours): Each student will do many hands-on activities in and out of the class lecture times. These will vary from simple coding to a student-selected project using Java/Python programming language. This project will require the transference of the concepts and skills of the course to an area of application of interest to the student.

Evaluation and Grading: The grading and grading scale will be based on the following tables:

- **Grading**: The evaluation will be based on the following criteria and weights.

Criteria	Weights (%)
Homework	40
Hands on Activities/Quizzes	20
Midterm	20
Final Exams	20

- **Grading Scale**

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F
%	100-93	92-90	89-87	86-83	82-80	79-77	76-73	72-70	69-67	66-63	62-60	59-

All final grades will be rounded up to the nearest whole number when assigning letter grades.

Attendance: Let me know (by email to babafemi.sorinolu@houghton.edu with the subject “CSCI 218 Attendance”) if and why you will be missing a class. In general, this should be done before the class you will miss.

Exams: There will be one midterm exam in the class and one cumulative final exam. Make-up exams will only be given in extenuating circumstances; prior arrangements must be made if possible.

Tentative dates for each exam are as follows:

- Midterm Exam 1: Tuesday, February 6th, 12:30–2:20pm
- Midterm Exam 2: Tuesday, March 19th, 12:30–2:20pm.
- Final Exam: Tuesday, May 7th, 8:00–10:00am.

Academic Honesty: Honesty is the foundation on which all intellectual endeavors rest. To use the ideas of others without acknowledging the authors of those ideas belies the nature and purpose of academic life. At Houghton, where we strive to live out Christian calling and commitment, personal integrity, including academic honesty, should be the hallmark of all our work and relationships. Students are expected to exhibit extreme care relative to personal honesty in all academic work, including in-class and out-of-class learning experiences, such as exams, quizzes, journals, papers, and research projects. Please refer to the catalog for the details of this policy.

Self-Reporting of Disabilities: If you have an academic or physical disability that requires accommodations, it is up to you to self-report any such disability to the office of Academic Support and Accessibility Services in the Center for Student Success located on the 1st floor of Chamberlain. (585-567-9622) With appropriate documentation, you will be afforded the necessary accommodation. For more information about the Center for Academic Success and Advising, go to:- https://www.houghton.edu/current-students/center-for-student-success/academic-support-and-accessibility-services/

Tentative Schedule: Below is a tentative calendar for the topics we will cover during each class period. These topics correspond to the chapters in our textbook.

Week	Class Date	Lecture Topics
1	1/9 - 1/11	Introduction to Java Tools and software for Java Writing the first code
2	1/16 - 1/18	Introduction to Java Classes and Objects Creating objects Introduction to Data types; Java Variables
3	1/23 - 1/25	Java Modifier types Using the Scanner Class Operators in Java
4	1/30 - 2/1	Control flow statements Introduction to Loops Concepts of OOP – Inheritance & Overriding
5	2/8	Polymorphism & Abstraction Data Structures in Java
6	2/13 - 2/15	Java Methods – Types of Methods Parameter Passing Exception handling in Java
7	2/20 - 2/22	Encapsulation Interfaces Constructors Concept of overloading
8	3/5 - 3/7	String Class; Number Class; Character Class; Multithreading in Java Introduction to Swing GUI Development
9	3/12 - 3/14	Swing GUI Development
10	3/21	Introduction to Spring boot
11	3/26	Introduction to Spring boot
12	4/2 - 4/4	Java Database Connectivity (JDBC)
13	4/9 - 4/11	Files and IO Streams
14	4/16 - 4/18	Networking and socket programming
15	4/23 - 4/25	**Review**
16	**5/7**	**Final Exams**